Nihar Gaddam

Software Development Engineer | Distributed Systems, Kubernetes, Cloud Infrastructure
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Summary

Software Development Engineer with 3 years of experience in cloud infrastructure and backend engineering, building distributed systems in Go on Kubernetes. Develops the Go-based Kubernetes operator (control plane) that manages the lifecycle of thousands of isolated, multi-tenant Splunk Cloud deployments at Cisco, with a focus on observability, disaster recovery, and security. Reclaimed 2,385 abandoned deployments through automated garbage collection and raised Azure network provisioning success from 33% to 98%.

Experience

Software Development Engineer | Cisco Systems Inc. | Pittsburgh, PA Aug 2024 – Present

- Designed and maintained features for a Go-based Kubernetes operator (control plane) managing the full lifecycle of thousands of hosted Splunk Cloud deployments (the isolated, multi-tenant product instances used by customers and internal teams), plus a customer-facing configuration API.
- Root-caused a cross-region disaster recovery (DR) ordering defect that let teardown delete a live primary while its standby replica was active; enforced dependency-safe ordering in the control loop so a primary is removed only after its replicas, preventing data loss and eliminating the resulting manual-cleanup tickets and on-call pages.
- Engineered an automated garbage-collection system for stale infrastructure, reclaiming 2,385 abandoned deployments and cutting unnecessary cloud spend fleet-wide in production alone.
- Authored a design proposal integrating a downstream pooling service with the control plane to pre-provision Azure subscriptions for deployment networking (otherwise created on demand), raising Azure network provisioning success from 33% to 98% and cutting provisioning time to about 20 min.
- Created the first metrics and Service Level Objective (SLO) observability layer for the platform (Prometheus, OpenTelemetry), instrumenting per-phase latency and success/failure metrics across every provisioning stage (infrastructure, software, and final delivery) with data-driven SLOs; shipped pageable alerts that catch genuine regressions proactively and cut cross-team mispages by 21%.
- Extended that layer with a dedicated OpenTelemetry collector sidecar for the provisioning operator, bringing teardown under the same measurement coverage as provisioning, and corrected phase-specific histogram buckets to eliminate inflated percentile estimates while holding metric-stream cost in check.
- Delivered a declarative, self-serve workflow to replace months-long manual provisioning of preview deployments (short-lived demo instances teams trial before adopting the official version), with validations that strictly isolate preview from production; scaled to about 100/month with zero manual coordination, accelerated customer testing workflows 38%, and drove adoption across five teams.
- Led a backward-compatible migration from PBKDF2 to HKDF (AES-GCM) across the shared cryptography library and the Kubernetes control plane, adding a versioned cipher codec, CLI and API support, automatic codec detection, and certificate-encryption contracts behind a version-gated rollout; verified bidirectional compatibility on staging deployments so existing and newly written ciphertext stay readable fleet-wide.
- Automated service-to-service authentication with ephemeral credentials, removing the need to raise manual tickets to update the latest build version, and drove encryption-at-rest to 100% coverage across 3,000+ deployments.
- Automated authentication-token rotation across 15 service accounts, eliminating manual intervention and improving token-management efficiency about 40%.
- Extended the customer-facing configuration API with IPv6 support and contributed to its CLI, broadening supported addressing and product capability for customers.
- Held 24/7 on-call across 5 Kubernetes clusters and 8 services plus the control plane, triaging and resolving production incidents on page while reducing technical debt through targeted refactors across development, staging, and production.
- Partnered directly with customers to reproduce and resolve production-level issues in both the control plane and the configuration API.

Application Development Associate | Accenture Solutions Pvt. Ltd. | Hyderabad, India Dec 2020 – Oct 2021

- Migrated client infrastructure from on-premises to Amazon Web Services (AWS) and Microsoft Azure, improving scalability and reducing operational costs by 20%; maintained development and production environments, reducing downtime by up to 40%.
- Executed high-risk production changes with minimal disruption, maintaining a 99.8% success rate in change management.
- Automated collection of performance, CPU, and disk-usage data across development and production servers with scheduled PowerShell jobs, replacing manual per-server checks.

- Developed an automated workflow to retrieve, cleanse, and organize data from Snowflake into Azure SQL, keeping datasets development-ready without manual preparation.
- Analyzed real-time server performance trends in Power BI, reducing response time to performance issues by 30%; managed Extract, Transform, Load (ETL) dataflows from Azure SQL, SharePoint, Redshift, MongoDB, and Snowflake into BI reporting pipelines.
- Cleaned and transformed raw Excel and Oracle data with Power Query, automating data-integration steps and improving data accuracy 22%.
- Stood up and maintained a Power BI data gateway bridging on-premises databases (Oracle, MySQL, PostgreSQL) to the reporting layer.
- Implemented an error-logging system in Power BI with custom error-log tables and automated capture and notification, enabling quick identification of data-refresh failures and reducing reporting downtime.

Projects

timberdb | *Go* Jun 2026

- Developed a time-partitioned, time-to-live (TTL) native log-structured merge (LSM) storage engine in Go; benchmarked 2.0x Pebble, 5.8x Badger, and 13x bbolt append throughput at 3 allocs/op, with 6.6x faster scans than Badger.

SVID-Exchange | *Go* Mar 2026

- Implemented a Zero Trust token exchange service that converts SPIFFE SVIDs into scoped, short-lived ES256 JSON Web Tokens (JWTs) over mutual TLS (mTLS), eliminating static shared secrets for service-to-service authentication.

Saga-Conductor | *Go* Mar 2026

- Created a lightweight saga-pattern orchestrator for distributed transactions, served over gRPC as a single binary with no external database: callers define each step with a forward action and a compensating action, and the conductor executes them in order, automatically rolling back completed steps in reverse on failure.

Skills

Languages: Go, Python, Rust, C++, Java
Cloud Platforms: Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform (GCP)
Containers and Orchestration: Docker, Kubernetes
CI/CD: GitHub Actions, Apache Airflow
Infrastructure as Code (IaC) and Configuration Management: Terraform, Puppet
Data Stores and Processing: MySQL, PostgreSQL, MongoDB, Kafka, Spark
Monitoring and Observability: Prometheus, Grafana, OpenTelemetry, Splunk, SignalFx

Education

Carnegie Mellon University Pittsburgh, PA
Master of Science in Computer Engineering (Applied Advanced Study) *Dec 2023*

- GPA: 3.75/4.00
- Relevant Coursework: Distributed Systems, Cloud Computing, Foundations of Computer Systems, Computer Vision, Machine Learning for Signal Processing, Introduction to Deep Learning

Sreenidhi Institute of Science and Technology Hyderabad, India
Bachelor of Technology in Electronics and Communication Engineering *Jun 2020*

- GPA: 9.39/10.00, Minor in Computer Science and Engineering

Certifications

Certified Kubernetes Application Developer (CKAD) Mar 2024
Cloud Native Computing Foundation (CNCF) *Expires Mar 2027*